5.8 FETAL CIRCULATION

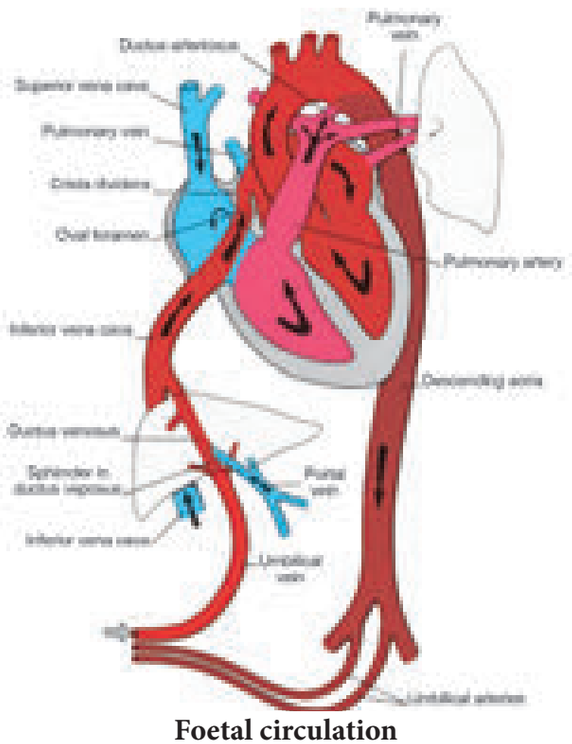
Definition: The circulation of oxygenated blood, de-oxygenated blood, nutritive material etc, in the foetus is termed as foetal circulation. The blood vessels responsible for foetal circulation are

1. **One Umblical Vein:** It carries the oxygenated blood from the placenta to the growing fetus.
2. **Two Umblical Arteries:** Both arteries carries all the de-oxygenated blood out of the fetus and carries de-oxygenated blood from the fetus to the placenta.

The shunts involved in foetal circulation

There are three shunts present in a fetus, they are:

1. **Ductus Venosus:** The Ductus Venosus shunts the portion of left umblical vein blood flow directly to the inferior vena cava
2. **Ductus Arteriosus:** It allows most of the blood from the right-ventricle to bypass the fetus' fluid-filled non-functioning lungs. Connects the pulmonary artery to the proximal descending aorta.
3. **Foramen Ovale**: It allows the blood to enter the left atrium from the right atrium It is an opening in the intra-atrial septum.



Foetal circulation

Step 1: The placenta accepts the blood without oxygen from the fetus through blood vessels that leave the fetus through the Umbilical Cord (Umblical Arteries).

Step 2: When blood goes through the placenta it picks up oxygen

Step 3: The oxygenated blood then returns to the fetus via the umbilical cord (umbilical vein).

Step 4: The oxygenated blood that enters the fetus passes through the fetal liver and enters the right atrium of the heart.

Step 5: Foramen Ovale allows the oxygenated blood to go from the right atrium to left atrium and then to the left ventricle and out the aorta. As a result the blood with the more oxygen gets in to the brain.

Step 6: Blood coming back from the fetus's body also enters the right atrium, but the fetus is able to send this deoxygenated blood from the right atrium to the right ventricle (the chamber that normally pumps blood to the lungs). Most of the blood that leaves the right ventricle in the fetus bypasses the lungs through the ductus arteriosus.

Step 7: The ductus arteriosus sends the deoxygenated blood to the organs in the lower half of the fetal body. This also allows for the deoxygenated blood to leave the fetus through the umbilical arteries and get back to the placenta to pick up oxygen.

XII_Nursing_Vocational_Unit 5.indd 155 1/11/2020 2:46:13 PM